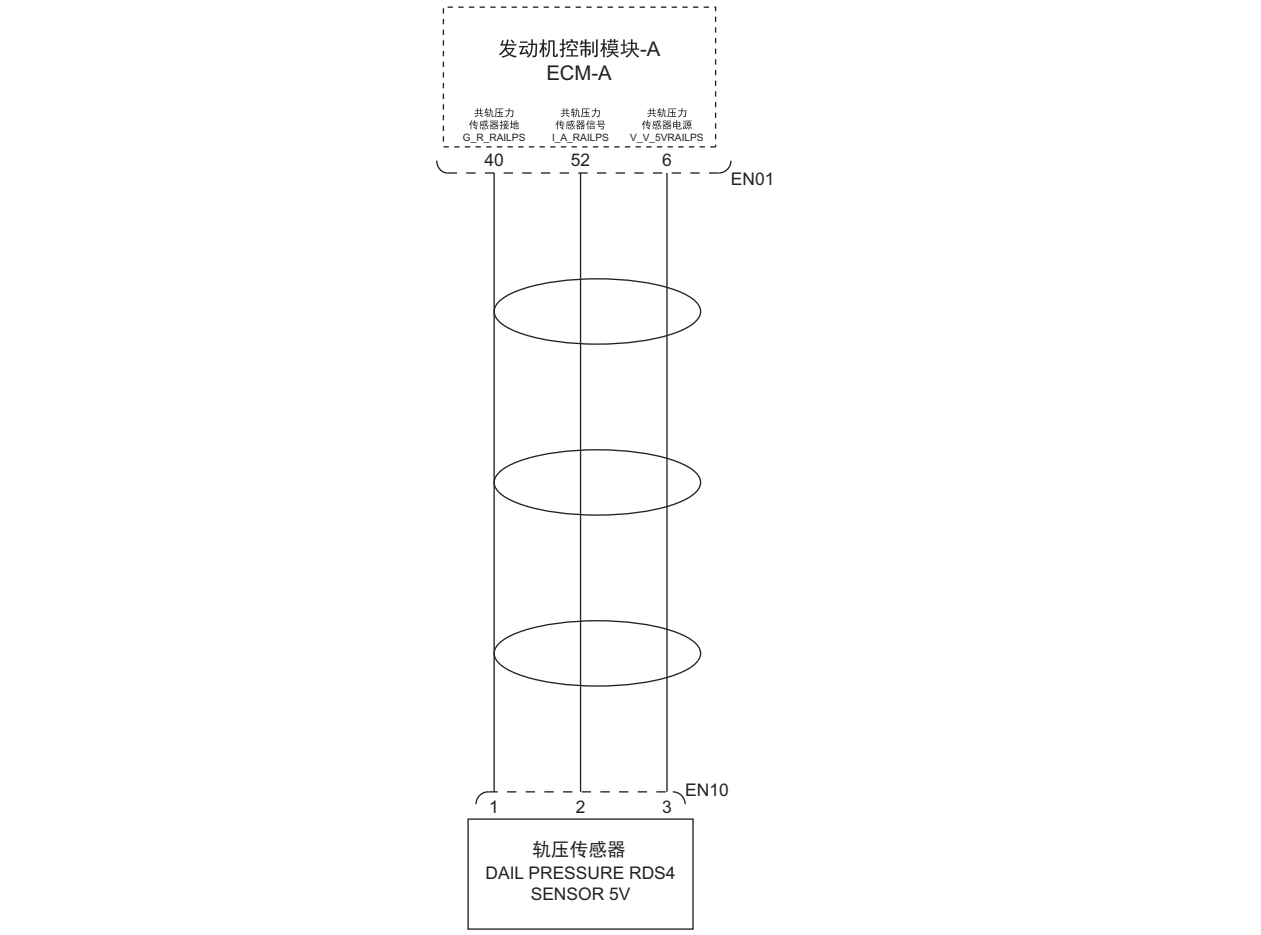
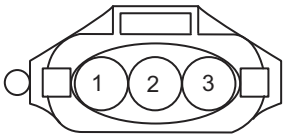
## P0087、 P0088

| Description of fault diagnosis codes |
| --- |
| P0087-00：Physical Range Check low for signal of intake air temperature sensor (after intercooler) |
| P0088-F0：Physical Range Check high for signal of intake air temperature sensor (after intercooler) |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • Fault in harness or harness plug<br>• Rail pressure sensor fault<br>• High-pressure oil pump fault<br>• Low-pressure oil pump fault<br>• System fault of low-pressure oil pump<br>• Fuel exhaustion |
| **Description of circuit diagram** |
| The rail pressure sensor is used to measure the actual pressure inside the high-pressure oil rail in the high-pressure common rail system. The rail pressure sensor is equipped with one 5v reference voltage circuit, one reference voltage circuit of low level and one signal circuit. The rail pressure sensor converts the pressure signal into voltage signal, and then sends the signal to engine control unit, so as to achieve closed-loop control of the fuel pressure. |

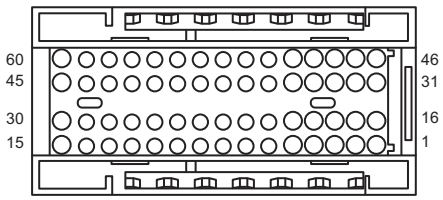
**Circuit diagram**




EN10

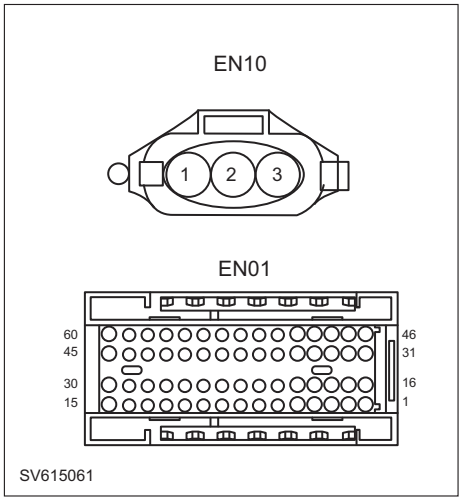



EN01




SV615025

| | | |
|---|---|---|
| **Troubleshooting procedures** | | |
| Step 1 | General inspection. | |
| | | 1 Check the fuse for damage.<br>2 Check battery capacity.<br>3 Check the engine control module and the harness plug of rail sensor for damage, poor contact, aging and looseness.<br>4 Check whether the result is normal. |
| | | **No →** Repair or replace the faulted parts. |
| **Yes ↓** | | |
| Step 2 | Check whether there is any fuel leakage or not. | |
| | | 1 Start the engine.<br>2 Check by visual inspection whether there is any leakage of fuel pump, fuel rail and fuel pipe.<br>3 Check whether the result is normal? |
| | | **No →** Repair or replace the faulted parts. |
| **Yes ↓** | | |
| Step 3 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Diagnose the engine control system with diagnostic apparatus.<br>4 Is there any other fault code except for P0087,P0088? |
| | | **Yes →** Reference:DTC Summary List (EU5). |
| **No ↓** | | |

| Step 4 | Check the circuit of rail pressure sensor(open circuit). |
|---|---|
|  | 1 Turn the ignition switch to OFF.<br>2 Disconnect the rail pressure sensor and the harness plug of engine control module<br>3 Measure the resistance of harness plug between rail pressure sensor and engine control module.<br>Measure circuit: the resistance between the EN10 plug terminal 1 and EN01 plug terminal 40.<br>Measure circuit: the resistance between EN10 plug terminal 2 and EN01 plug terminal 52.<br>Measure circuit: the resistance between EN10 plug terminal 3 and EN01 plug terminal 6.<br>**Standard value: ≈ 0Ω**<br>4 Check whether the result is normal? |
| | **No →** Repair or replace the faulted parts. |
| **Yes ↓** | |
| Step 5 | Check the circuit of rail pressure sensor(short circuit). |
| | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug of rail pressure sensor.<br>3 Measure the voltage between the harness plug of rail pressure sensor and ground.<br>Measure circuit: the voltage between EN10 plug terminal 1 and ground.<br>Measure circuit: the voltage between EN10 plug terminal 2 and ground.<br>Measure circuit: the voltage between EN10 plug terminal 3 and ground.<br>**Standard value: ≈ 0V**<br>4 Measure the resistance between the harness plug of rail pressure sensor and ground.<br>Measure circuit: the resistance between EN10 plug terminal 1 and ground.<br>Measure circuit: the resistance between EN10 plug terminal 2 and ground.<br>Measure circuit: the resistance between EN10 plug terminal 3 and ground.<br>**Standard value: ∞**<br>5 Check whether the result is normal? |
| | **No →** Repair or replace the faulted parts. |
| **Next ↓** | |
| Step 6 | Complete the check and confirm whether the system works normally nor not. |